

ETABS

Software Course | Duration: 7 Weeks

A complete 3D building analysis and design course using ETABS — covering seismic and wind analysis, storey drift and irregularity checks, and RC/steel frame design for multi-storey structures.

Who Should Enroll: Structural engineers, design trainees, and STAAD.Pro users transitioning into 3D building analysis for mid-to-high-rise projects.

Prerequisites

- Basic structural analysis and RC design concepts
- STAAD.Pro exposure is helpful but not required

Learning Outcomes

- Build a complete 3D building model with correct storey and diaphragm data
- Apply gravity, wind, and seismic loads per Indian codes
- Run response spectrum and equivalent static seismic analysis
- Check storey drift, base shear, and structural irregularities
- Design RC/steel frame members and produce a design report

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Model Setup	– Interface, grids, and storey data – Modeling beams, columns, slabs, and shear walls
Week 2	Properties & Sections	– Material and section property libraries – Assigning releases, offsets, and rigid diaphragms
Week 3–4	Loading	– Gravity, wind, and seismic load application – Load combinations per IS 875 & IS 1893 – Diaphragm behavior: rigid vs semi-rigid modeling
Week 5	Seismic Analysis	– Response spectrum and equivalent static analysis – Storey drift, base shear, and irregularity checks
Week 6	Design Modules	– Concrete and steel frame design within ETABS – Cross-checking design output against manual calculations
Week 7	Capstone Project	– Full 3D seismic analysis and design of a multi-storey RC building – Final design report and performance summary

Capstone Project

Complete 3D seismic analysis, design, and performance evaluation of a multi-storey RC building.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.